

An Enhanced Multi Layer Neural Network To Detect Early Cardiac Arrests

**Ranga Swamy
Sirisati**

Associate Professor,
Department of CSE,
Vignan's Institute of
Management and
Technology for
Women, Kondapur,
Ghatkesar, Hyderabad
, Telangana, E-Mail:
sirisatiranga@gmail.com

**C. Srinivasa
Kumar**

Professor, Department
of CSE, Vignan's
Institute of
Management and
Technology for
Women, Kondapur,
Ghatkesar, Hyderabad
, Telangana, E-Mail:
4drskumar41@gmail.com

**A. Gautami
Latha**

Professor, Department
of CSE, Sridevi
Women's Engineering
College,
Vattinagulapally,
Hyderabad. 500075,
Telangana, India, E-
Mail:
gauthamilatha@gmail.com

**B. Narendra
Kumar**

Professor, Department
of CSE, Sridevi
Women's Engineering
College,
Vattinagulapally,
Hyderabad. 500075,
Telangana, India, E-
Mail:
bnkphd@gmail.com

**Kanusu
Srinivasa Rao**

Assistant Professor,
Department of
Computer Science and
Technology, Yogi
Vemana University,
Kadapa, india, EMail:
kanususrinivas@gmail.com

Abstract- Patients and doctors are paying increasing attention to health-care automation because it can save a person's life by predicting ailments early. Many people are suffering from chronological diseases as a result of altering eating habits, regardless of age or gender. "Heart Attacks" is a severe ailment that requires attention from time to time. To date, all automated systems have built models using either classical or ensemble machine learning techniques. Overfitting has affected only a few of these systems, such as random forest and SVM algorithms. As a result, the proposed approach has chosen the "Multi Layer Preceptron" neural network technique, which solves the problem of overfitting and generates an accurate number of correct labels linked with the training model. Instead of using all of the variables mentioned in the dataset, the suggested method assists clinicians in predicting a heart attack in a user at an early stage by assessing only 7 top informative attributes. The model was also compared to other classifiers in order to establish the state of the art, which was determined to be "97.23 percent."

Keywords: Feature Engineering, Greedy Genetic Algorithm, 8- Layered Multi Layer Preceptron Architecture, Ensemble Algorithm, Meta Classifier

I. INTRODUCTION:

Classification is type of supervised machine learning algorithm, which defines a mapping function to draw a relationship between set of inputs and class labels defined in the dataset. The types of machine learning algorithms are shown in figure 1.

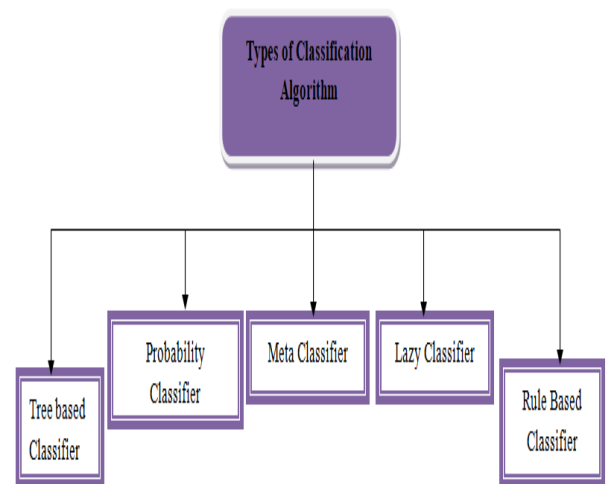


Figure 1: Categories of Classification Algorithm

In Tree based classifier, the proposed system has chosen J48 Algorithm to identify whether a person suffer from heart attack or not by constructing tree based on entropy as decision parameter. Among all the trees, J48 is considered as best decision tree algorithm because it evaluates all the possible subsets and generates the tree as shown in figure 2.

Conditional independence is important in determining the impact of attributes on the class label parameters in a probability classifier. The Bayesian probability, which is popular for generating precise values based on decision rules, is used in conditional probability. The suggested system used a naive Bayesian approach, which yielded positive outcomes during the prediction phase. The

analysis techniques using the popular data mining tool known as “WEKA”. The WEKA tool is a simple data experimental and visualize tool with good GUI representation for the business analyst [10].

3.1. Exploratory Data Analysis:

TABLE 1: META DATA ABOUT HEART DISEASE DATASET

S. No	Name	Description	Possible Values	Minimum	Maximum	Mean	Standard Deviation
1	Age	It represents the age of person. It ranges from 29 to 77	41 different values	29	77	54.366	9.082
2	Sex	It represents the gender of the person	{0,1} 0: female 1: male	0	1	0.683	0.466
3	CP	It represents the chest pain based on the location, cause and relief	{0,1,2,3} 0: All conditions are satisfied 1: Any two conditions are satisfied 2. At least one condition is satisfied 3. none of the conditions are satisfied	0	3	0.967	1.032
4	trestbps	It measures the level of blood pressure in resting mode in mmHg units	49 different values	94	200	131.64	17.538

5	Chol	It measures cholesterol in mg/dL	152 different values	126	564	246.264	51.83
6	Fbs	It measures sugar level in blood during fasting	{0,1} 0: True 1: False	0	1	0.149	0.356
7	restecg	It represents ECG state in resting mode	{0,1,2} 0: Normal 1: Abnormality 2: Probably yes	0	2	0.528	0.526
8	thalach	Measures the heart beat rate	91 different values	71	202	149.647	22.905
9	Exang	Include exercise at location	{0,1} 0: No 1: Yes	0	1	0.327	0.47
10	oldpeak	Depression caused due to exercise in units of mm	40 different values	0	6.2	1.04	1.61
11	Slope	Slope value obtained at peak point of exercise	{0,1,2} 0: Upslope 1: Flat Slope 2: Down Slope	0	2	1.399	0.616
12	CA	Number of major vessels affected	{0,1,2,3,4}	0	4	0.729	1.023
13	Thal	Measures the portion that is unable to detect thallium	{0,1,2,3} 0: normal 1: fixed 2: reversible 3:	0	3	2.314	0.612

			Abnor mal				
14	Tar get	Class Label	{No, Yes}	-	-	-	-

It is observed that except the class label, remaining all the attributes are numerical. So, the system has performed computation of mean and standard deviation [11]. The data analysis results are represented in figure 3.

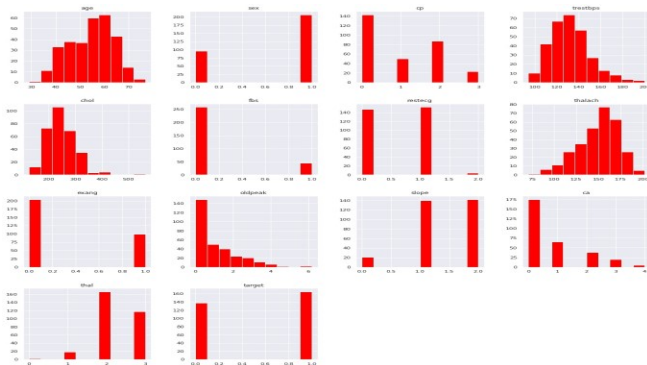


Figure 3: Data Analysis on Attributes

Inferences from Analysis: Pre-processing is the basic and efficient step in building the accurate model. To fill the missing values with either mean or median or mode, the proposed system tries to infer the following points and choose the appropriate method to clean the data [12] [13].

- Age, chol, thalach attributes are processed using the mode
- trestbps, old peak and CA are filled using the median
- Remaining values are cleaned using the mean value of the particular attribute.

IV. PROPOSED METHODOLOGY

The proposed system based on the features selected from the genetic algorithm, it implements the multi layer perceptron. Multiple layers are added to the neural network and a bias value to classify the patients. To minimize the error rate, the weights are adjusted using equation (1)

$$New_{weight} = old_{weight} + l_{rate} * (original_value - predicted_value) + b$$

ias – (1)

The designed model identifies a static learning rate, l_{rate} which is measurement to define the adaptability of the model to new environment, in which it is undergoing training. The difference between the original value and predicted value is considered as the error rate. A constant value known as “bias” is added to normalize the values of input. An 8- layered neural network is designed in which one layer act as “input”, one layer act as “output” and

remaining 6 layers acts as “hidden layers”. All these layers are defined with dense functionality, in which every input is passed through every unit of every layer. The MLP architecture is famous for its composition of activation function that performs affine transformation to convert the complex activities into simple linear ones. So that accurate systems can be build with in logarithmic time. The overall architecture of the model is represented in figure 4.

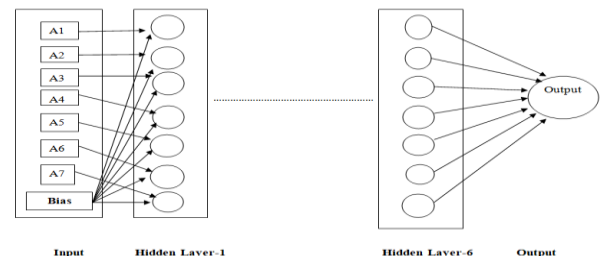


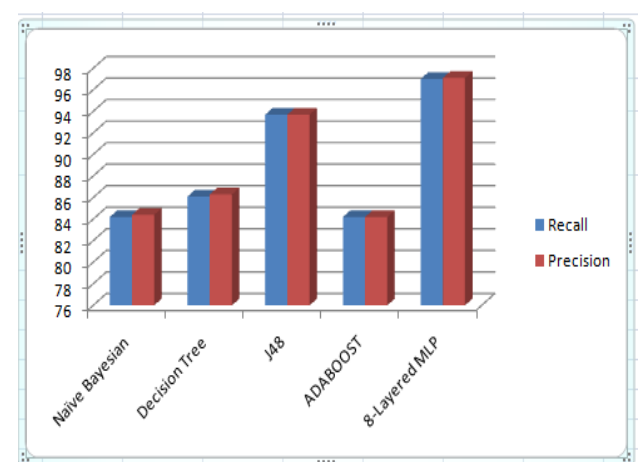
Figure 4: 8-Layered Architecture of MLP

V. EXPERIMENTAL RESULTS

The proposed algorithm has compared the previous models mentioned in the literature survey and presented in table 2.

TABLE 2: COMPARITIVE ANALYSIS ON RESULTS OBTAINED

Algorith m	Accur acy	Rec all	Precisi on	F1- Scoe re	AU C	Kap pa
Naïve Bayesian	84.1	84.2	84.4	84.0	90.0	67.7
Decision Tree	86.1	86.1	86.3	86.1	92.6	71.8
J48	93.7	93.7	93.7	93.7	97.2	87.35
ADABOOST	84.1	84.2	84.2	84.2	93.1	68.1
8-Layered MLP	97.02	97.0	97.1	97.0	97.8	94.0



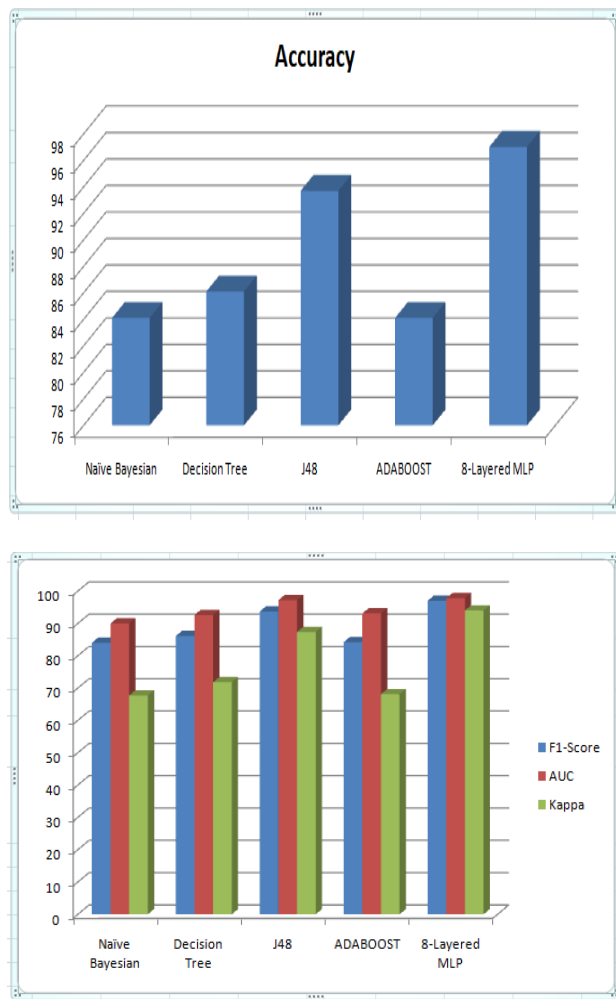


Figure 5(a) (b) (c): Comparative Results on Evaluation Metrics

From figure 5(a) (b) (c), it is a clear view that all the metrics of evaluation are good in terms of the proposed algorithm, and the second-best model can be considered as J48, which is a type of tree-based classifier. On X-axis, the evaluation metrics are mentioned, and on the y-axis, the percentage of measurement with an interval of 20 units is mentioned.

VI. CONCLUSION

Death Rates are increasing with cardiac arrest year by year from the past few years irrespective of age and gender. Early prediction of heart attacks can save a life of a human being. For early prediction of heart attack, an automated model integrated with ANN techniques is proposed in this paper, with an accuracy of 97.02%. The main advantage of this model is that it considers the impact of each attribute on the class label, and based on the current instance in the testing data, it adjusts the weight parameters. The time to build this model is a bit high when compared to the ensemble algorithms. So in future work, researchers can identify the dynamic activation functions and can be embedded in the hyper tuned layers.

REFERENCES

- [1]. Kumar, N. K., Sindhu, G. S., Prashanthi, D. K., & Sulthana, A. S. (2020, March). Analysis and Prediction of Cardio Vascular Disease using Machine Learning Classifiers. 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS). 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS). <https://doi.org/10.1109/icaccs48705.2020.9074183>.
- [2]. Motarwar, P., Duraphe, A., Suganya, G., & Premalatha, M. (2020, February). Cognitive Approach for Heart Disease Prediction using Machine Learning. 2020 International Conference on Emerging Trends in Information Technology and Engineering (Ic-ETITE). 2020 International Conference on Emerging Trends in Information Technology and Engineering (ic-ETITE). <https://doi.org/10.1109/ic-etite47903.2020.242>.
- [3]. Barik S, Mohanty S, Rout D, Mohanty S, Patra A.K., Mishra A.K. (2020) Heart Disease Prediction Using Machine Learning Techniques. In: Pradhan G, Morris S, Nayak N. (eds) Advances in Electrical Control and Signal Systems. Lecture Notes in Electrical Engineering, vol 665. Springer, Singapore. https://doi.org/10.1007/978-981-15-5262-5_67.
- [4]. R. SwamySirisati, M. S. Rao and S. Thonukunuri, "Analysis of Hybrid Fusion-Neural Filter Approach to detect Brain Tumor," 2020 Sixth International Conference on Parallel, Distributed and Grid Computing (PDGC), 2020, pp. 460-464, doi: 10.1109/PDGC50313.2020.9315809.
- [5]. Saw, M., Saxena, T., Kaithwas, S., Yadav, R., & Lal, N. (2020, January). Estimation of Prediction for Getting Heart Disease Using Logistic Regression Model of Machine Learning. 2020 International Conference on Computer Communication and Informatics (ICCCI). 2020 International Conference on Computer Communication and Informatics (ICCCI). <https://doi.org/10.1109/iccci48352.2020.9104210>.
- [6]. Wang, M., Yao, X., & Chen, Y. (2021). An Imbalanced-Data Processing Algorithm for the Prediction of Heart Attack in Stroke Patients. IEEE Access, 9, 25394–25404. <https://doi.org/10.1109/access.2021.3057693>.
- [7]. Balasubramaniam, Vivekanadam. "Artificial Intelligence Algorithm with SVM Classification using Dermoscopic Images for Melanoma Diagnosis." Journal of Artificial Intelligence and Capsule Networks 3, no. 1 (2021): 34-42. 2.
- [8]. Dhaya, R. "Deep net model for detection of covid-19 using radiographs based on roc analysis." Journal of Innovative Image Processing (JIIP) 2, no. 03 (2020): 135-140.
- [9]. Swamy, s. ranga, c. srinivasa kumar, and a. gauthami latha. "an efficient skin cancer prognosis strategy using deep learning techniques.
- [10]. Diwakar, M., Tripathi, A., Joshi, K., Memoria, M., Singh, P., & kumar, N. (2021). Latest trends on heart disease prediction using machine learning and image fusion. Materials Today: Proceedings, 37, 3213–3218. <https://doi.org/10.1016/j.matpr.2020.09.078>
- [11]. Swamy, S. Ranga, et al. "Dimensionality reduction using machine learning and big data technologies." Int. J. Innov. Technol. Explor. Eng. (IJITEE) 9.2 (2019): 1740-1745.
- [12]. A. M. Reddy, K. SubbaReddy and V. V. Krishna, "Classification of child and adulthood using GLCM based on diagonal LBP," 2015 International Conference on Applied and Theoretical Computing and Communication Technology (iCATccT), Davangere, 2015, pp. 857-861, doi: 10.1109/ICATCCT.2015.7457003.
- [13]. Ayaluri MR, K. SR, Konda SR, Chidirla SR. 2021. Efficient steganalysis using convolutional auto encoder network to ensure original image quality. PeerJ Computer Science 7:e356 <https://doi.org/10.7717/peerj-cs.356>